Library Management System — Final Report

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1. Introduction

The purpose of this project was to develop a complete Library Management System using Java and object-oriented programming concepts. The system allows users to manage library items, borrow and return materials, search for items, and generate reports.

2. Project Objectives

• Create a functional library management system
• Apply object-oriented programming principles
• Implement inheritance and polymorphism
• Use abstract classes and interfaces
• Implement recursive and stream-based searching
• Use Java collections such as List, Stack, Queue, Set, and Map
• Implement exception handling
• Load and save data using CSV files
• Create sorting functionality
• Perform unit testing using JUnit

3. Technologies and Tools Used

• Java
• IntelliJ IDEA
• Maven
• JUnit 5
• Lombok
• CSV File I/O
• Visual Paradigm
• GitHub

4. System Overview

The Library Management System was designed around two abstract classes: User and Item. Student, Teacher, and Admin inherit from User while Book, DVD, and Magazine inherit from Item.

5. Object-Oriented Programming Concepts

The project demonstrates abstraction, inheritance, polymorphism, and interfaces using Java class hierarchies and interfaces such as Reportable.

6. Collections Used

• List
• Stack
• Queue
• Set
• Map

7. Searching Features

Recursive and stream-based searching were implemented with case-insensitive matching and duplicate filtering.

8. Borrowing and Returning System

The system validates borrowing limits, item availability, and student restrictions while using exception handling to manage invalid operations.

9. File Persistence Using CSV

CSV files are used to load and save users and items.

10. Validation

ISBN validation ensures ISBN numbers contain exactly 13 digits.

11. Sorting Features

Items and users can be sorted alphabetically using Java Comparators.

12. Unit Testing

JUnit tests verify borrowing, returning, searching, sorting, and validation functionality.

13. Challenges Faced

Challenges included UML design, CSV parsing, duplicate filtering, and implementing borrowing validation.

14. Conclusion

The project successfully implemented all required features while demonstrating major Java programming concepts.